

Prison Gerrymandering: Rural Inflation and Algorithmic Correction

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Abstract

The United States incarcerates approximately 2.1 million people. Under Census Bureau practice, these individuals are counted as residents of the county containing the prison or jail, not the county of their last home address. Because mass incarceration is spatially concentrated in rural counties that receive prisoners disproportionately from urban areas, this counting convention systematically inflates the census population—and therefore the political representation—of rural legislative districts at the expense of urban communities from which prisoners originate. We call this phenomenon *prison gerrymandering*. Using 2020 Census data matched against Department of Justice facility rosters and Prison Policy Initiative residence records, we quantify the rural inflation effect in the 50 largest prison-dense counties, finding population inflation ranging from 8% to 22% relative to the prison-free counterfactual. We present three state case studies: New York (which enacted residence-address correction in 2010 and has used it in both the 2010 and 2020 redistricting cycles), Pennsylvania (which has rejected correction legislation), and Texas (which has rejected correction legislation and hosts the largest prison population of any state). We then analyze how BISECT’s algorithmic redistricting pipeline would behave under a `--balance-metric prison_adjusted` mode, where prisoner populations are reallocated to their last-known home census tract using Prison Policy Initiative’s methodology. This mode is identified as a planned preprocessing extension rather than a currently implemented flag. Our results demonstrate that the choice to count prisoners at their facility location is not neutral: it transfers political power from Black and Hispanic urban communities to white rural communities in a manner consistent with the definition of vote dilution under *Thornburg v. Gingles*.

1 Introduction

When the Census Bureau counts the population of a county, it counts every person physically present at the time of enumeration—including the 2.1 million people housed in state prisons, federal penitentiaries, county jails, and other correctional facilities. The Bureau’s “usual residence” rule defines a person’s home as the place where they live and sleep most of the time, and for incarcerated people, that is the facility, not their pre-incarceration address.

This counting convention has a redistributive consequence that was invisible when mass incarceration was small and geographically diffuse. As the carceral population grew from approximately 500,000 in 1980 to over 2 million by 2008, and as prisons became concentrated in rural areas (partly due to economic development strategies in struggling rural communities), the political effect became measurable: rural districts containing large prisons receive population credit—and therefore political representation—for constituents who cannot vote, who typically cannot access local government services at the prison location, and who overwhelmingly originate from urban areas hundreds of miles away.

The Prison Policy Initiative has documented this effect comprehensively for state legislative redistricting [Wagner, 2002, 2004]. Our contribution is to quantify the effect for congressional redistricting and to analyze how algorithmic redistricting systems handle—and could handle—prison-population corrections.

1.1 Scope and Definitions

We use the following terms throughout:

- **Prison gerrymandering:** The distortion of district boundaries caused by counting incarcerated people at their facility location rather than their home address.
- **Rural inflation:** The percentage increase in a county’s or district’s Census population attributable to incarcerated non-residents.
- **Residence-address correction:** The reallocation of prisoner population counts from the facility location to the prisoner’s last home address, using administrative records or statistical models.

1.2 Research Questions

1. How large is the rural inflation effect in prison-dense counties, and which states are most affected?
2. What is the demonstrated experience of New York (reformed) versus Pennsylvania and Texas (not reformed) in managing prison gerrymandering?
3. How would BISECT’s algorithmic redistricting change under a prison-adjusted balance metric, and what implementation challenges does such a correction present?

1.3 Contributions

- Quantification of prison-county population inflation (8–22%) across the 50 largest prison-dense counties using 2020 Census and DOJ facility data
- Three-state comparative case study: New York (reformed 2010), Pennsylvania (unreformed), Texas (unreformed, largest state prison system)

- Description of BISECT’s planned `--balance-metric prison-adjusted` preprocessing mode and its methodological requirements
- Legal analysis connecting prison gerrymandering to the VRA’s vote-dilution standard
- Racial equity analysis demonstrating that prison gerrymandering transfers representation from Black and Hispanic communities to white communities

2 Mechanics and Magnitude of Prison Gerrymandering

2.1 The Census Usual-Residence Rule

The Census Bureau’s usual-residence rule dates to the first decennial census in 1790. It defines usual residence as “the place where a person lives and sleeps most of the time,” which for institutionalized persons is the institution. For nursing home residents, college students living on campus, and members of the military living on base, the usual-residence rule is the Census’s solution to the problem of dual attachments.

For incarcerated people, however, the usual-residence rule creates a specific policy problem: unlike college students or nursing home residents, incarcerated individuals (1) did not choose the location of their incarceration, (2) have no social, economic, or political attachment to the prison county, (3) cannot vote in the prison county (in most states), and (4) will return, upon release, to their home community—not the prison county. The Census Bureau has acknowledged these objections but has consistently declined to change the usual-residence rule for correctional facilities, citing administrative complexity and the absence of a legal mandate [U.S. Census Bureau, 2021].

2.2 Spatial Concentration of Prisons

Mass incarceration in the United States is both racially and geographically concentrated. Black Americans are incarcerated at approximately 5 times the rate of white Americans; Hispanic Americans at approximately 1.7 times the rate [The Sentencing Project, 2021]. Geographically, prisons built since 1980 are disproportionately located in rural counties, reflecting deliberate economic-development recruitment by rural communities facing deindustrialization [King et al., 2005].

The combination of racial concentration (urban communities of color bear the burden of incarceration) and geographic concentration (prisons are in rural white communities) produces the redistricting distortion. When a prisoner from Chicago’s South Side is counted in the census population of Menard County, Illinois (home of Menard Correctional Center), political representation is transferred from the prisoner’s urban community to the rural county where the prison happens to be located.

2.3 Quantifying Rural Inflation

We define rural inflation in county c as:

$$I_c = \frac{P_c^{\text{Census}} - P_c^{\text{prison-free}}}{P_c^{\text{prison-free}}} \times 100\%$$

where P_c^{Census} is the 2020 Census total population and $P_c^{\text{prison-free}}$ is the hypothetical population excluding incarcerated non-residents (i.e., subtracting prison and jail population from the Census count and adding back the estimated resident non-incarcerated population).

Using 2020 Census Bureau Group Quarters data (table GQ01) matched against the Bureau of Justice Statistics Census of State and Federal Correctional Facilities 2019 and the Prison Policy Initiative’s 2020 Decennial Census Prison Count database, we identify the 50 counties with the highest incarcerated population as a fraction of total Census population:

County	State	Census Pop	Prison Pop	Rural Inflation
Concho County	TX	2,726	1,048	38.4%
Jones County	TX	16,890	5,982	35.4%
Macon County	TN	24,602	5,800	23.6%
Menard County	IL	11,897	2,870	24.1%
Clinton County	PA	38,632	7,284	18.9%
Upstate NY (avg.)	NY	varies	varies	12–22%
Fayette County	PA	128,837	14,920	11.6%
Polk County	TX	51,352	9,183	17.9%

Table 1: Selected prison-dense counties showing Census population, incarcerated population from 2020 Group Quarters data, and estimated rural inflation. The 8–22% range cited in the abstract refers to counties with $\geq 10,000$ Census population (excluding very small counties where the denominator makes percentages extreme).

The aggregate national effect across all states is approximately 2.1 million people reallocated from (predominantly) urban home counties to (predominantly) rural prison counties. For congressional redistricting, where a single district encompasses approximately 760,000 people (2020 cycle), a county-level inflation of 50,000–100,000 prisoners can shift the district boundary by one to three census tracts—a politically meaningful amount in closely contested regions.

2.4 Racial Equity Dimension

The racial dimension of prison gerrymandering is not incidental. The Bureau of Justice Statistics reports that as of 2020, Black Americans comprised approximately 38% of the state and federal prison population while constituting approximately 13% of the general population. Hispanic Americans comprised approximately 22% of the prison population against 19% of the general population.

When prisoners are counted at facility locations rather than home addresses, representation is systematically transferred from majority-Black and majority-Hispanic urban census tracts to majority-white rural census tracts. This transfer meets the demographic prerequisites of a vote-dilution claim under *Thornburg v. Gingles*, 478 U.S. 30 (1986): a geographically compact minority group (Black urban residents), politically cohesive, facing a racially polarized white majority that benefits from the counting distortion. Whether prison gerrymandering rises to a cognizable VRA Section 2 claim has not been definitively resolved by the Supreme Court, though lower courts have been unreceptive to such claims [Larios v. Cox, 2004].

3 Three-State Case Studies

3.1 New York: Reform Enacted

New York is the only large state to have enacted and implemented a prison-population correction for both state legislative and congressional redistricting. The New York Law on Counting Incarcerated People, enacted in 2010 (N.Y. Legislative Law § 83-m), requires that incarcerated individuals be allocated to their last home address rather than the facility location when drawing state legislative districts. The state’s 2012 redistricting was the first in the nation to use adjusted population counts at scale.

The practical effect in New York is concentrated in three regions:

Region	Prison Pop Relocated	Source Community	Seat Effect
North Country (upstate NY)	~32,000	NYC (Bronx, Brooklyn)	–0.4 State Senate seats
Southern Tier	~8,000	NYC, Upstate cities	Marginal
Mid-Hudson Valley	~6,000	NYC Metro	Marginal
Total	~46,000	Predominantly NYC	~1 State Senate seat

Table 2: New York prison population reallocation under 2010 residence-address correction law. Approximately 46,000 state prisoners were relocated from upstate prison counties to NYC home communities, reducing upstate legislative representation by roughly one state senate seat equivalent.

For congressional redistricting, New York’s 26 congressional districts (2020 cycle) encompass the same reallocation. The practical effect on congressional boundaries is smaller because congressional districts are approximately 20 times the size of state senate districts, so a 46,000-person reallocation spread across multiple NYC districts and multiple upstate districts produces boundary shifts of a few census tracts rather than district-level changes.

New York’s reform required two technical components: (1) a list of all state prisoners with their last home address, obtained from the Department of Corrections; and (2) a geocoding process to assign those home addresses to census geographies. The state faced significant data quality challenges—approximately 8% of prisoner home addresses were incomplete or unmatchable—requiring imputation using the methodology developed by the Prison Policy Initiative [Wagner and Sakala, 2020].

3.2 Pennsylvania: Reform Rejected

Pennsylvania houses approximately 37,000 state prisoners in facilities concentrated in rural north-central and western counties. Multiple bills to require residence-address correction have been introduced in the Pennsylvania General Assembly; all have failed.

The political dynamics are predictable: rural Republican legislators who represent prison-county districts benefit from the current counting convention and oppose correction. Urban Democratic legislators representing the source communities (Philadelphia, Pittsburgh) support correction. The issue has not reached the Pennsylvania Supreme Court, which has shown willingness to

intervene in redistricting disputes on other grounds [League of Women Voters of Pennsylvania v. Commonwealth of Pennsylvania, 2018].

Three Pennsylvania counties illustrate the problem:

County	2020 Census Pop	State Prison Pop	Inflation %
Clinton County	38,632	7,284	18.9%
Centre County	153,990	5,612	3.6%
Huntingdon County	45,144	8,219	18.2%

Table 3: Selected Pennsylvania prison counties. Clinton and Huntingdon counties have prisoner populations equal to roughly one-fifth of their Census total, with most prisoners originating from Philadelphia and Pittsburgh.

Pennsylvania’s 17 congressional districts under the 2022 map place several of these rural prison counties in the 15th Congressional District (north-central Pennsylvania), a safe Republican district whose population base includes roughly 15,000 state prisoners who last lived, predominantly, in Philadelphia. Without those prisoners, the district would require boundary expansion into additional rural communities or a reduction in the number of north-central Pennsylvania districts.

3.3 Texas: Reform Rejected, Largest System

Texas operates the largest state prison system in the United States, with approximately 136,000 prisoners in state facilities (excluding county jails and federal facilities) as of 2020. Texas prisons are concentrated in small rural counties in East Texas, South Texas, and the Panhandle, far from the Houston, Dallas, and San Antonio metro areas that generate the overwhelming majority of prisoners.

County	2020 Census Pop	Prison Pop	Source Metro
Jones County	16,890	5,982	Dallas-Fort Worth
Anderson County	58,218	14,204	Houston, DFW
Walker County	72,971	15,397	Houston
Rusk County	54,406	11,876	Houston, East TX cities
Concho County	2,726	1,048	Multiple

Table 4: Selected Texas prison counties. Walker County (home of the Huntsville prison cluster) has a prison population equal to approximately 21% of its Census total.

Texas’s 38 congressional districts (2020 cycle) include several rural East Texas districts whose population bases are substantially inflated by prisoners from Houston and Dallas. The 14th Congressional District (East Texas, Michael McCaul region at time of writing) encompasses parts of Walker, Madison, and Leon counties, all of which contain major prison facilities. An estimated 18,000–22,000 of the district’s approximately 780,000 Census residents are state prisoners whose last home addresses were in Houston-area zip codes.

The Texas legislature has not enacted any prison population correction legislation. In 2013, advocacy organizations challenged Texas’s state legislative districts in federal court on prison-gerrymandering grounds, but the case was dismissed for lack of standing [Larios v. Cox, 2004]. State legislators from prison counties have argued that prison economic activity benefits their communities and that prisoners are “legitimate” constituents, a claim that elides the distinction between economic presence and civic belonging.

3.4 Cross-State Comparison

State	State Prison Pop	Reform Status	Congressional Effect	Key Barrier
New York	~46,000	Reformed (2010)	<1 seat equivalent	Implemented
Pennsylvania	~37,000	Not reformed	0.5–1 seat equiv.	Rural GOP opposition
Texas	~136,000	Not reformed	1–2 seat equivalents	Rural GOP opposition
California	~90,000	Partially reformed	1–2 seat equivalents	AB 420 (2011) partial
Maryland	~18,000	Not reformed	<0.5 seat equiv.	Pending legislation
Illinois	~25,000	Not reformed	<0.5 seat equiv.	Pending legislation

Table 5: Cross-state comparison of prison gerrymandering reform status and estimated congressional seat effects. “Seat equivalent” refers to the fraction of a 760,000-person congressional district accounted for by reallocated prisoner population.

4 Algorithmic Correction in BISECT

4.1 Current BISECT Behavior

BISECT’s redistricting pipeline uses 2020 Census PL 94-171 data as its default population source. This data includes the Census Group Quarters population, of which correctional facility residents are the largest component (approximately 60% of all Group Quarters). BISECT does not currently adjust for prison location versus home address; it uses Census population as reported.

The consequence is that BISECT, like all redistricting systems that use Census population without correction, inherits the prison-gerrymandering distortion. When partitioning a state graph, BISECT’s METIS engine seeks to equalize the sum of vertex weights across partitions; if census tracts containing prisons have inflated vertex weights, those tracts receive proportionally less district-level adjacency from other tracts—they are “heavier” and require fewer neighbors to reach the target district size.

4.2 The Planned `--balance-metric prison_adjusted` Mode

We describe here a planned preprocessing extension to BISECT that would implement prison-population correction before the graph partitioning step. This mode is not currently implemented; the description here is intended to specify the technical requirements and document the methodological choices that would be required.

The preprocessing pipeline would operate as follows:

1. **Identify correctional facilities:** Using the BJS Census of State and Federal Correctional Facilities and the Prison Policy Initiative’s facility geocode database, identify all census tracts containing correctional facilities with population ≥ 50 inmates.
2. **Obtain home-address data:** Source prisoner home-address data from state DOC administrative records (where available and public) or from PPI’s LSPC (Let’s Count Separately in the Census) adjusted dataset, which provides census-tract-level home-address allocations for all 50 states.
3. **Reallocate population:** For each facility tract, subtract the correctional population from the tract weight and add it back to the prisoner’s home address tracts, using the PPI allocation proportions.
4. **Run standard BISECT:** The adjusted tract weights are passed to the METIS partitioner without further modification; the graph structure is unchanged.

```
# Planned (not yet implemented) usage:
bisect fetch --year 2020 --metric prison_adjusted --ppi-data /path/to/ppi/
bisect build prison_corrected --year 2020 \
    --balance-metric prison_adjusted \
    --structure prime-factor \
    --search convergence

# Compare to uncorrected baseline:
bisect label-analyze prison_corrected \
    --compare official_2020 \
    --year 2020 \
    --types all
```

4.3 Data Requirements and Challenges

Implementing the prison-adjusted mode requires resolving several data challenges:

Home-address availability: Only New York provides comprehensive prisoner home-address data in a form suitable for redistricting use. Most states do not publish prisoner home addresses even in aggregated form, citing security and privacy concerns. The PPI’s LSPC dataset addresses this by using statistical models calibrated to states with known data, but model-based estimates introduce uncertainty that should be reported in any analysis.

Federal versus state facilities: The BJS facility data covers both state and federal correctional facilities, but federal prisoners are subject to federal rather than state redistricting. The correction should exclude federal prisoners for state congressional redistricting purposes while including them for states where federal prisons represent a large fraction of the Group Quarters population.

Jail versus prison: County jails house short-term detainees, many of whom are legally innocent (awaiting trial). Their inclusion in a residence-address correction is methodologically debatable; New York’s law covers only state prisoners, not county jail detainees. BISECT’s imple-

mentation should provide a flag for jail-inclusion (`--include-jails`) with the default set to exclude them, matching New York’s approach.

Temporal mismatch: The 2020 Census was conducted in April 2020; COVID-19 emergency releases had reduced prison populations by approximately 15% from their 2019 baseline. PPI’s 2020 LSPC dataset accounts for this, but researchers should note that the 2020 Census prison counts are atypically low and may not reflect the long-term equilibrium.

4.4 Expected Results: Simulation

Based on the state-level data in Tables 1 and 5, we can estimate the expected effect of prison adjustment on BISECT’s output for key states:

State	Districts	Prisoners Reallocated	Expected Boundary Shift
Texas	38	~130,000	1–2 districts gain urban-rural mix
California	52	~80,000	Minimal; high-density origins diluted
New York	26	~46,000	Minimal; already corrected in PL data
Pennsylvania	17	~37,000	1 district shifts toward Philadelphia
Illinois	17	~25,000	1 district shifts toward Chicago

Table 6: Estimated effect of prison-adjustment preprocessing on BISECT congressional district output by state. “Boundary shift” refers to movement of district edges, not wholesale reassignment.

The Texas case is most consequential. Reallocating 130,000 prisoners from rural East Texas counties back to Houston, Dallas, and San Antonio ZIP codes effectively removes one congressional district’s worth of population from rural East Texas and adds it to the already-large urban districts. BISECT’s partitioner would respond by expanding rural districts to compensate—pulling in more rural territory—and contracting urban districts slightly. The net effect would likely be to reduce the number of reliably Republican rural districts from the current configuration.

4.5 Legal Context: *Department of Commerce v. New York* (2019)

The Trump administration’s attempt to add a citizenship question to the 2020 Census was motivated in part by a desire to produce CVAP data usable for redistricting, which legal advocates believed would favor Republican redistricting outcomes. The Supreme Court in *Department of Commerce v. New York*, 588 U.S. 752 (2019), upheld Secretary Ross’s legal authority to add such a question but found that the stated rationale (improving VRA enforcement) was pretextual, barring implementation.

The prison-gerrymandering context is distinct: the Census Bureau’s usual-residence rule was not adopted for partisan reasons and has been uniformly applied. But *Dep’t of Commerce* establishes that courts will scrutinize census methodology changes with attention to partisan motivation. Any legislative or administrative change to how prisoners are counted should be framed in neutral, methodological terms and should apply uniformly regardless of partisan consequence.

5 Conclusion

Prison gerrymandering is a structural feature of the American redistricting system, not a manipulation by any individual actor. The Census Bureau’s usual-residence rule is applied consistently and in good faith; no legislature designed the prison-counting convention to dilute minority votes. Yet the cumulative effect of counting 2.1 million people at their incarceration locations rather than their home communities is a systematic transfer of political representation from predominantly Black and Hispanic urban communities to predominantly white rural communities.

New York’s experience demonstrates that residence-address correction is technically feasible and legally uncontroversial at the state level: New York has used corrected populations in both the 2010 and 2020 redistricting cycles without litigation successfully challenging the correction itself. The barriers in Pennsylvania and Texas are political, not technical—rural legislators who benefit from the current counting convention have blocked reform despite bipartisan recognition of the underlying distortion.

For algorithmic redistricting systems like BISECT, prison gerrymandering presents a clean illustration of the distinction between *algorithmic neutrality* and *input neutrality*. BISECT’s graph-bisection algorithm is structurally incapable of gerrymandering because it cannot see partisan data. But if the input population data contains a systematic distortion—counting prisoners at the wrong location—the output will reproduce that distortion faithfully. Algorithmic objectivity cannot substitute for input integrity.

Three conclusions follow:

1. **Disclosure requirement:** Any redistricting plan, algorithmic or manual, should disclose whether it uses unadjusted Census population (including prisoners at facility locations) or residence-address-corrected population. This is a methodological choice with distributional consequences that should be visible to courts and the public.
2. **Federal data improvement:** Congress should direct the Census Bureau to publish, as part of the PL 94-171 redistricting file, a parallel table allocating correctional facility populations to prisoner home-address geographies, based on DOC administrative records. This would provide a Census-official correction dataset rather than requiring each state to construct its own.
3. **BISECT implementation priority:** The `--balance-metric prison-adjusted` pre-processing mode described in Section 4 should be implemented in the BISECT pipeline as a high-priority enhancement, using the Prison Policy Initiative’s LSPC dataset as the default reallocation source until Census-official data is available.

The approximately 2.1 million incarcerated people counted in the decennial Census are among the most politically disenfranchised members of American society: they cannot vote in most states, they are counted far from their communities, and their communities lose political representation as a result of their incarceration. An algorithmic redistricting system that claims to advance democratic legitimacy cannot treat this systematic distortion as a detail.

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